

SIAM AREFIN

[Email](#) | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

Research Interests: Machine Learning, Computer Vision, Generative AI, Large Language Models, Multimodal AI, Medical/Biological AI, AI Systems

Research Experiences

omicML: An Integrative Tool of Bioinformatics and Machine Learning Algorithms to Identify Transcriptional Biomarkers

Dec 2024 – Oct 2025

Machine Learning Research Engineer | Collaboration with Department of Biochemistry and Molecular Biology (BMB), SUST

- Implemented scalable machine learning deployment and monitoring pipelines using MLOps best practices.
- Collaborated with bioinformatics researchers to integrate computational modeling with biological large-data analysis.
- Worked on an interdisciplinary research problem connecting machine learning with transcriptomic biomarker identification.

Research-Oriented Industry Experience

Kite Games Studio

Nov 2025 – Jan 2026

Software Engineer (ML Team) Intern | Mohakhali DOHS, Dhaka — Onsite

- Conducted R&D in image processing, machine learning, and generative AI for production-oriented applications.
- Worked with Diffusion and FLUX-based generative AI models for image generation and processing.
- Contributed to an Android application with 5M+ downloads and 700K+ daily active users.

Academic Credentials

Shahjalal University of Science and Technology (SUST), Bangladesh

Mar 2022 – Jul 2026

Bachelor of Science in Software Engineering

CGPA: 3.10/4.00

Selected Research Projects

HealthMonitor – AI-Powered Health Monitoring App

- Developed and deployed a MERN-stack application with a custom-trained machine learning model for cardiovascular disease prediction.
- Integrated the Google Gemini API for intelligent advice and developed a RAG-based system for personalized insights using medical records and health-related data.
- Implemented secure authentication using JWT.

AI Processing API – AI/ML Backend System

- Designed a modular AI backend using layered architecture, Provider/Adapter pattern, and Dependency Injection.
- Built REST APIs for audio transcription and medical-document OCR with validation, structured extraction, testing, and Dockerized deployment.

Academic and Competitive Achievements

- **1st Place** – Multi-Instance Object Detection Challenge, hosted by Duality AI (a USA startup).
- **1st Runner-up** – ML Olympiad, CO₂ Emissions Prediction Challenge.
- **4th Place** – DL Enigma 1.0, SUST CSE Carnival 2024.
- **Top 10** – DL Sprint 3.0, Bengali AI Math Olympiad.
- Solved **800+ problems** on Codeforces and other competitive programming platforms.

Technical Skills

Machine Learning:	Machine Learning, Deep Learning, NLP, CNNs, RNNs, Transformers, Scikit-learn, PyTorch, TensorFlow, Feature Engineering
Generative AI / LLMs:	LLMs, Prompt Engineering, Fine-tuning, RAG, Vector Databases, LangChain, AI Agents, Hugging Face, Seq2Seq, Tokenization, Diffusion Models, Google Gemini API
Computer Vision:	OpenCV, YOLO, Object Detection, Image Processing and Generation, Ensemble Learning
Programming:	Python, C, C++, Java, JavaScript
AI Systems / MLOps:	FastAPI, REST APIs, Docker, Git/GitHub, CI/CD, Azure, SQL, MongoDB
Software Engineering:	OOP, SOLID, Design Patterns, DSA, Software Architecture, Unit Testing

Research Direction

Developing robust and deployable machine learning systems for computer vision, generative AI, large language models, multimodal learning, and interdisciplinary biomedical/bioinformatics applications.